Java Loops Intensive Practice Before Arrays

These exercises are designed to make loops feel natural. Practice slowly, trace variables manually, and focus on understanding iteration.

Student Instructions

#	Instruction
1	Push every exercise to GitHub.
2	Create one Java file per exercise.
3	Do NOT use Artificial Intelligence tools.
4	Predict output before running programs.
5	Trace every loop manually.
6	Debug your own errors first.
7	Use meaningful variable names.
8	Practice consistently every day.
9	Do not copy from classmates.
10	Focus on understanding, not memorization.

FOR LOOP PRACTICE

#	Exercise
1	Print numbers from 1 to 50.
2	Print numbers from 50 down to 1.
3	Print all even numbers between 1 and 100.
4	Print all odd numbers between 1 and 100.
5	Print multiples of 5 between 1 and 100.
6	Print the square of numbers from 1 to 20.
7	Print the cube of numbers from 1 to 10.
8	Find the sum of numbers from 1 to 100.
9	Find the product of numbers from 1 to 10.
10	Print the multiplication table of 9.
11	Print all numbers divisible by both 3 and 5.

12	Print numbers from 1 to 100 skipping multiples of 4.
13	Print numbers between 1 and 100 divisible by 7.
14	Print numbers from 20 down to 1.
15	Print the first 50 natural numbers.

WHILE LOOP PRACTICE

#	Exercise
16	Print numbers from 1 to 30 using a while loop.
17	Print numbers from 30 down to 1.
18	Print all even numbers from 2 to 50.
19	Find the sum of numbers from 1 to 50.
20	Print the multiplication table of 12.
21	Ask the user for 5 numbers and calculate the total.
22	Keep asking the user for numbers until they enter 0.
23	Create a password checker using a while loop.
24	Count backwards from 100 to 1.
25	Print all multiples of 8 between 1 and 200.
26	Find the factorial of 5 using a while loop.
27	Find the largest number from user inputs.
28	Find the smallest number from user inputs.
29	Print numbers skipping multiples of 6.
30	Count how many numbers exist between 1 and 100.

DO WHILE PRACTICE

#	Exercise
31	Print numbers from 1 to 10 using a do while loop.
32	Print numbers from 10 down to 1.
33	Print all even numbers from 2 to 40.
34	Print all odd numbers from 1 to 41.
35	Print the multiplication table of 5.
36	Keep asking the user for a positive number.
37	Create a simple login simulation.
38	Display a menu until the user exits.
39	Ask the user for grades until they enter -1.

40	Find the sum of numbers entered by the user.
41	Create a mini ATM simulation.
42	Count the number of user attempts before success.
43	Print numbers divisible by 9 between 1 and 200.
44	Print a countdown timer from 20 to 0.
45	Build a simple calculator using do while.

NESTED LOOP PRACTICE

#	Exercise
46	Print a square star pattern.
47	Print a right triangle star pattern.
48	Print an inverted triangle pattern.
49	Print a pyramid pattern.
50	Print a rectangle pattern.
51	Print a hollow square pattern.
52	Print a multiplication table from 1 to 10.
53	Print coordinates using nested loops.
54	Print numbers in rows and columns.
55	Print a diamond star pattern.

Arrays will become easier when loops become natural. Master iteration, tracing, updating variables, and nested loops first.